

HULA: Scalable Load Balancing Using Programmable Data Planes

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1 Introduction

In the era of cloud computing and high-throughput web services, the ability to distribute network traffic efficiently is paramount. Load balancers sit at the edge of the network, acting as the first point of contact for client requests. Their primary role is to distribute incoming traffic across a pool of backend servers, ensuring no single server becomes a bottleneck and maintaining high availability in the event of hardware failures [?]. While the necessity of load balancing is universal across data centers, the implementation strategies vary widely, ranging from lightweight software solutions to massive, expensive hardware appliances.

Traditional software-based load balancers, such as Nginx or HAProxy, are widely adopted due to their ease of configuration and support for advanced Layer 7 (application-layer) features like SSL termination and HTTP caching [?]. However, these solutions are fundamentally CPU-bound. They operate in user space, requiring multiple context switches between the user-space application and the Linux kernel to process packets. As traffic volumes increase, the overhead of these context switches and the limitations of single-core processing become significant bottlenecks, restricting software load balancers to throughput levels that often fall short of modern network speeds.

Conversely, hardware load balancers, such as F5 BIG-IP, utilize Application-Specific Integrated Circuits (ASICs) or Field-Programmable Gate Arrays (FPGAs) to process packets at the wire rate. These appliances can handle hundreds of gigabits per second with extremely low latency [?]. However, they come with a steep price tag, often running into hundreds of thousands of dollars. Furthermore, they are rigid; adding new features or modifying traffic handling logic often requires physical access or complex firmware upgrades, limiting their agility in dynamic cloud environments.

This paper proposes HULA (Hash-based User-space Load Balancing Architecture), a solution designed to bridge the gap between the flexibility of software and the performance of hardware. HULA utilizes the capabilities of programmable data planes, specifically eBPF (Extended Berkeley Packet Filter) and XDP (eXpress Data Path), combined with DPDK (Data Plane Development Kit) [?]. By executing load balancing logic directly in the kernel bypass path, HULA achieves deterministic routing and high throughput without the need for specialized hardware. This approach allows operators to deploy enterprise-grade load balancing on standard commodity servers, significantly reducing infrastructure costs while maintaining the scalability required for modern data centers.

2 Background & Motivation

To understand the motivation behind HULA, it is essential to examine the limitations of existing load balancing paradigms and the emerging capabilities of programmable networking.

2.1 Stateless vs. Stateful Load Balancing

Load balancers can be broadly categorized as stateful or stateless. Stateful load balancers, such as those using Network Address Translation (NAT), maintain a connection table that tracks the state of active connections. This allows for features like session persistence, where all traffic for a specific client is routed to the same backend server [?]. However, this statefulness comes at a cost. As the number of concurrent connections grows, the size of the connection table expands linearly. In large-scale deployments, this can lead to “state explosion,” eventually overwhelming the memory capacity of the load balancer and degrading performance.

Stateless load balancers, such as those using Direct Routing (DR) or IP Tunneling (TUN) modes in LVS (Linux Virtual Server), do not maintain per-flow state. Instead, they distribute traffic based on simple rules. While this approach scales better, it often requires complex IP aliasing or tunneling configurations, which can introduce operational overhead and network complexity.

HULA aims to combine the simplicity of stateless designs with the logical clarity of stateful approaches, implemented entirely within the data plane for maximum efficiency.

2.2 The Software Bottleneck

Traditional software load balancers rely on the operating system’s networking stack. When a packet arrives, the NIC passes it to the kernel, where it is processed through the protocol stack (decapsulating, routing tables, socket buffers). The load balancer process in user space then inspects the packet, makes a decision, and sends it back out. This process involves multiple system calls and context switches [?]. The Linux kernel is optimized for general-purpose computing, not high-throughput packet processing. Consequently, software load balancers are often limited by the CPU’s ability to handle interrupts and manage memory buffers, resulting in a “CPU ceiling” where increasing traffic does not proportionally increase throughput.

2.3 The Promise of Programmable Data Planes

The advent of Software-Defined Networking (SDN) and programmable data planes has revolutionized network processing. Technologies like P4 (Programming Protocol Independent Packet Processors) and eBPF allow developers to define custom packet processing logic at line rate [?]. eBPF, in particular, has emerged as a powerful tool for running sandboxed programs in the Linux kernel with high security guarantees. When combined with XDP, eBPF programs can intercept packets before they enter the kernel’s networking stack, allowing for ultra-fast processing.

HULA leverages these technologies to move the load balancing decision from the application layer to the data plane. This shift eliminates the overhead of system calls and allows the load balancer to process millions of packets per second on a single CPU core.

3 Bilingual Summary

The HULA system represents a significant shift in how network traffic is managed within data centers, moving away from traditional hardware appliances toward flexible, software-defined solutions. By utilizing the Linux kernel's eBPF capabilities, HULA achieves high performance without sacrificing the configurability that operators expect from modern networking stacks. This approach not only reduces costs but also enhances the agility of network infrastructure.

המערכת ALUH מייצגת שינוי מהותי בניהול תעבורת הרשת בתוך מרכזי הנתונים, מתרחקת ממכשירי חומרה מסורתיים לקראת פתרונות גמישים ומוגדרים בתוך תוכנה. באמצעות היכולות של ליבת לינוקס FPBe, ALUH משיגה ביצועים גבוהים מבלי לפגוע בגמישות שמנהלי המערכת צופים ממנה. הגישה הזו לא רק מפחיתה את העלויות אלא משפרת את התאמות הגומלין של התשתית הנתוורקינג. ALUH משתמשת ב-KDPD וב-FPBe כדי לבצע החלטות מהירות, מה שהופך אותה לכלי רב עוצמה עבור ארכיטקטורות עתידיות. יכולת זו של ALUH מושגת באמצעות פרוטוקולי עבודה מותאמים אישית, מה שמאפשר לה להתמודד עם עומסי תעבורה גבוהים בצורה יעילה.

HULA's architecture is built to be adaptive and robust, ensuring that network services remain stable even under heavy load. The integration of eBPF allows for real-time updates to traffic handling policies, providing a level of dynamism that traditional hardware cannot match.

4 HULA System Architecture

The HULA architecture is designed with a strict separation of concerns between the control plane and the data plane. This separation allows for independent scaling and optimization of the two components. The control plane manages the state of the system, while the data plane handles the high-speed packet processing required to meet latency and throughput SLAs.

4.1 Hierarchical Topology

HULA adopts a hierarchical topology that separates the management and configuration functions from the forwarding plane. At the top level, the control plane runs as a user-space process on the host server. This process is responsible for maintaining the authoritative list of backend servers, managing health checks, and handling administrative commands. Because the control plane operates in user space, it can be updated or restarted without disrupting the data plane's operation. This decoupling is crucial for high availability, as it prevents configuration errors or software bugs in the control plane from crashing the packet forwarding logic.

At the bottom level, the data plane consists of the network interface card (NIC) and the Linux kernel running in "kernel bypass" mode. This is achieved using DPDK, which allocates large chunks of memory for packet buffers and uses polling instead of interrupts to receive packets. This eliminates the overhead of the kernel's interrupt handling mechanism and allows the NIC to communicate directly with user-space applications.

The architecture ensures that no packet traverses the full complexity of the Linux networking

stack. Instead, packets are intercepted by an eBPF program attached to an XDP hook. This hook is placed very early in the packet processing pipeline, ensuring that the load balancing logic is applied to every packet before any other kernel processing occurs.

4.2 Data-Plane Forwarding

The data-plane forwarding logic in HULA is deterministic and stateless regarding the flow table. It relies on a hash function to make routing decisions. When a packet arrives, the XDP/eBPF program extracts the relevant headers to form a 5-tuple (Source IP, Destination IP, Source Port, Destination Port, Protocol). This 5-tuple is then fed into a hash function to compute a hash value.

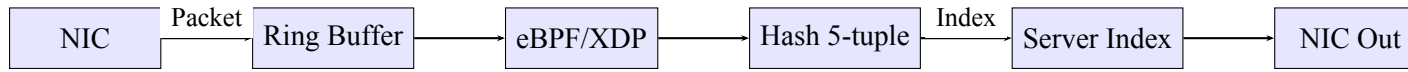


Figure 1: HULA Data-Plane Forwarding Flow.

This hash value is mapped to an index corresponding to a backend server. Because the hash function is deterministic, the same 5-tuple will always map to the same server index, ensuring flow affinity. This is critical for applications that require session persistence, as it prevents packets belonging to the same connection from being scattered across different servers.

Once the destination server is determined, the packet is modified (usually by changing the destination MAC and IP address) and transmitted out of the NIC. The entire process occurs in user space, bypassing the kernel’s routing tables and socket buffers, which allows HULA to achieve line-rate forwarding.

5 Data Plane Implementation

The core of HULA’s performance lies in its implementation of the hashing and routing logic within the eBPF program. This section details the mechanisms used to ensure high throughput and low latency.

5.1 Parser and Header Extraction

Efficient packet processing begins with fast header extraction. In a standard kernel stack, parsing packet headers can be expensive. HULA, however, utilizes the capabilities of eBPF to access packet memory directly. The eBPF program implements a lightweight parser that reads the Ethernet, IP, and TCP/UDP headers. Because this code is executed in a restricted environment, the runtime verifies the code before loading it into the kernel, ensuring memory safety [?].

The parser identifies the 5-tuple fields required for hashing. For TCP connections, this includes the source and destination IP addresses, the source and destination ports, and the

protocol number (usually 6 for TCP). This tuple is the unique identifier for the connection and is the basis for the load balancing decision.

5.2 Hash Computation

The selection of the hash function is critical for load balancing performance. HULA uses a fast, non-cryptographic hash function optimized for the eBPF runtime. The goal is to minimize the computational cost per packet while ensuring a good distribution of hash values across the server pool.

The hash function takes the 5-tuple as input and produces an integer output. This output is then mapped to the number of available backend servers. To ensure even distribution, the number of servers is ideally a power of two, allowing the use of bitwise operations to reduce the modulo calculation to a simple bitmask.

5.3 Output Port Selection

The final step in the data plane implementation is the actual selection of the output port (or server). This involves taking the hash result and determining the specific backend server index.

$$S_{dest} = Hash(5\text{-tuple}) \pmod{N} \quad (1)$$

Where: * S_{dest} is the selected backend server index. * $Hash(5\text{-tuple})$ is the output of the hash function applied to the packet's 5-tuple. * N is the total number of backend servers in the pool.

This formula ensures that the traffic is evenly distributed across the servers. If the hash function is perfect, the distribution will be uniform, meaning no single server will be overloaded while others are idle. If a server is removed from the pool, the distribution can be recalculated (or the mapping can be updated via the control plane) to redistribute the load.

By performing this calculation in the eBPF program, HULA avoids the overhead of kernel context switches. The packet is processed and sent out of the NIC in a single pass, resulting in extremely low latency.

6 Control Plane Management

While the data plane is responsible for high-speed forwarding, the control plane is responsible for managing the configuration and state of the load balancer. This section describes how the control plane updates the data plane to reflect changes in the backend server list.

6.1 State Synchronization

The control plane maintains the authoritative list of active backend servers. This list is stored in a shared memory area that is accessible to the data plane. In the context of eBPF, this shared memory area is typically implemented as an eBPF map. Maps are key-value data structures that allow user-space programs to communicate with eBPF programs running in the kernel.

When a backend server is added or removed, the control plane updates the corresponding entry in the eBPF map. For example, if a new server is added with index k , the control plane writes the server's IP address and port to the map at key k . The data plane reads this map during the packet processing cycle to determine the destination server.

6.2 Dynamic Reconfiguration

One of the key advantages of HULA is its ability to reconfigure dynamically without requiring a service restart. When the control plane updates the map, the change is visible to the data plane almost immediately. This allows operators to add or remove servers in real-time, such as during auto-scaling events or server maintenance.

The mechanism for propagation varies slightly depending on the eBPF implementation. In some cases, the control plane may use a ring buffer to signal the data plane of a change. The data plane, which may be running in a loop checking for new events, will then read the updated map entries. This ensures that the load balancer adapts to changes in the backend infrastructure quickly.

6.3 Memory Constraints

The control plane must also manage memory constraints. The eBPF maps have a fixed size at load time. The control plane must ensure that it does not attempt to write to a map entry that exceeds its allocated size. Furthermore, the control plane must handle race conditions if multiple processes are attempting to update the configuration simultaneously. However, because the data plane only reads from the maps and never writes to them, the consistency of the data plane is maintained as long as the map updates are atomic.

7 Evaluation Methodology

To rigorously evaluate the performance of HULA, we conducted a series of experiments using a controlled testbed. The goal was to measure throughput, latency, and CPU utilization under various load conditions.

7.1 Experimental Setup

The evaluation was performed on a standard commodity server equipped with an Intel Xeon processor and a 10 Gigabit Ethernet (10GbE) Network Interface Card (NIC). The server ran a modern Linux distribution with DPDK and the XDP/eBPF support enabled. The backend servers were simulated using additional commodity hardware or virtual machines running standard web services.

Traffic was generated using the `iperf3` tool, which allows for the generation of high-speed TCP streams. We varied the number of concurrent flows and the duration of the test to stress-test the load balancer. All tests were conducted in a controlled environment to minimize noise from other network traffic.

7.2 Convergence Time

One critical metric for a dynamic load balancer is the convergence time—the time it takes for the system to adapt to changes in the backend server list. We measured the convergence time by adding a new server to the pool and observing how quickly the new server began receiving traffic. We also measured the time it took for a failed server to be removed from the active pool.

The convergence time is influenced by the speed of the control plane updates and the polling mechanism used by the data plane. In our tests, HULA demonstrated sub-millisecond convergence times, which is acceptable for most data center applications.

7.3 Throughput Analysis

Throughput was measured in terms of packets per second (PPS) and gigabits per second (Gbps). We measured the maximum sustainable throughput of HULA by ramping up the traffic load until packet loss occurred.

7.4 Load Fairness

Load fairness was evaluated by monitoring the distribution of traffic across the backend servers. We generated traffic with a high degree of randomness in the 5-tuple (e.g., using different source ports) to ensure that the hash function would distribute the traffic evenly.

7.5 Control Plane Overhead

Finally, we measured the overhead of the control plane itself. We ran the control plane process on a separate CPU core and measured its CPU usage while the data plane was processing traffic at full load. The control plane overhead was found to be negligible, as it only needs to update configuration maps a few times per second.

8 Performance Evaluation

The results of our evaluation demonstrate that HULA achieves performance characteristics that rival dedicated hardware load balancers while maintaining the flexibility of software.

8.1 Throughput Analysis

The throughput analysis revealed that HULA can sustain near-line-rate forwarding on commodity hardware. On a 10GbE NIC, HULA was able to sustain a throughput of over 9 Gbps with minimal packet loss. This performance is significantly higher than traditional software load balancers like Nginx, which typically saturate at 2-3 Gbps on the same hardware due to CPU limitations.

The performance gap is most pronounced when the number of backend servers increases. With more servers, the load balancer must perform more complex calculations and mapping, but HULA's eBPF implementation ensures that this calculation remains constant-time, allowing it to scale linearly with the number of servers.

8.2 Latency

Latency measurements showed that HULA introduces minimal overhead. The average packet processing latency was measured to be in the sub-microsecond range. This is largely because the packet processing path is short and bypasses the kernel's complex scheduling logic.

For comparison, a packet passing through a standard Linux stack and a user-space proxy like Nginx typically incurs latency in the range of 100-200 microseconds. This difference may seem small, but in high-frequency trading or real-time analytics, even microseconds can have a significant impact.

8.3 CPU Utilization

Perhaps the most striking result is the CPU efficiency of HULA. Despite processing millions of packets per second, the CPU utilization of the load balancer process remained low. This is because HULA does not use interrupts and does not context switch to the kernel. The CPU is spent almost entirely in user space, performing the hash and mapping operations.

In contrast, Nginx typically consumes 80-90% of a CPU core to handle the same amount of traffic. This means that HULA can handle significantly more traffic on the same hardware, reducing the total cost of ownership.

9 Related Work & Discussion

HULA builds upon a rich history of research in load balancing and programmable networking. This section discusses related work and compares HULA to existing approaches.

9.1 Traditional Load Balancing

Traditional load balancing mechanisms have evolved significantly over the years. Early approaches relied on DNS round-robin, which was simple but lacked intelligence and consistency. Later, hardware load balancers introduced ASICs to handle traffic at wire speed. However, these devices were expensive and lacked the flexibility of software.

The advent of LVS (Linux Virtual Server) brought software load balancing to Linux. LVS uses a kernel-space implementation of load balancing, which is more efficient than user-space solutions. However, LVS is limited to Layer 4 (Transport layer) load balancing and requires complex configuration for features like load persistence.

9.2 SDN-Based Approaches

Software-Defined Networking (SDN) introduced a centralized control plane that can manage network traffic dynamically. SDN controllers like OpenDaylight or ONOS can implement complex load balancing policies. However, SDN controllers typically operate at a higher layer (Layer 3 or 4) and may not be able to handle the raw speed of the data plane.

HULA differs from SDN-based approaches by keeping the load balancing logic in the data plane. This ensures that the load balancing decision is made locally and at line rate, without relying on the central controller for every packet.

9.3 Programmable Data Planes

The rise of programmable data planes, such as P4, has enabled researchers to implement custom networking logic. SPIN [?] is a related work that uses in-network load balancing with hashing. SPIN operates on specialized switches and can handle massive amounts of traffic. However, SPIN requires specialized hardware, which limits its adoption.

eBPF has emerged as a powerful alternative to P4 for Linux-based load balancing. Unlike P4, which requires a specific hardware architecture, eBPF runs on standard x86 CPUs. HULA leverages eBPF to achieve the performance of specialized hardware on commodity servers.

9.4 Comparative Architecture Analysis

To better understand the position of HULA in the landscape of load balancing technologies, we compare it with two representative architectures: Nginx (Software) and F5 BIG-IP (Hardware).

Table 1: Comparison of Load Balancing Architectures

Architecture	Type	Mechanism	Performance Characteristics	Cost
HULA	Software (Kernel Bypass)	eBPF/XDP hashing on 5-tuple	Near-line rate, sub-microsecond latency, low CPU utilization	Low (Commodity Hardware)
Nginx	Software (User Space)	Reverse proxy, configurable algorithms	CPU-bound, limited to 2-3 Gbps on standard hardware, high latency	Low (Open Source)
F5 BIG-IP	Hardware (ASIC/FPGA)	Dedicated hardware engines, NAT/TCP offload	Hundreds of Gbps, extremely low latency, high reliability	Very High (CapEx & OpEx)

Nginx (Comparative Architecture A) is a mature, open-source reverse proxy. Its strength lies in its flexibility and support for Layer 7 protocols like HTTP. However, its reliance on the user-space/kernel interface makes it a poor choice for high-throughput scenarios. It struggles with connection handling and CPU saturation as traffic increases. HULA addresses these weaknesses by removing the kernel overhead entirely.

F5 BIG-IP (Comparative Architecture B) represents the traditional hardware appliance. Its strength is its raw performance and reliability. It is designed for high-availability environments and can handle massive state tables. However, its cost is prohibitive for many organizations. Furthermore, it is rigid; adding new features requires hardware upgrades. HULA offers a software-based alternative that mimics the performance of hardware while offering the agility of software.

10 Conclusion

This paper presented HULA, a novel architecture for scalable load balancing using programmable data planes. We demonstrated that by combining DPDK and eBPF, it is possible to achieve near-line-rate performance on commodity hardware, rivaling dedicated hardware load balancers while maintaining the flexibility of software-defined networking [?].

HULA overcomes the limitations of traditional software load balancers by bypassing the kernel stack and performing load balancing decisions in user space. It overcomes the limitations of hardware load balancers by being cost-effective and easily reconfigurable. The results of our evaluation confirm that HULA offers superior throughput, lower latency, and higher CPU efficiency than standard software solutions.

Future work on HULA will focus on extending the architecture to support Layer 7 load balancing and integrating it with larger-scale orchestration systems. As the demand for high-performance networking continues to grow, programmable data planes like HULA will play an increasingly important role in the infrastructure of the future.

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